



The real and financial effects of internal liquidity: Evidence from the Tax Cuts and Jobs Act[☆]

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ABSTRACT

The Tax Cuts and Jobs Act unlocked as much as \$1.7 trillion of U.S. multinationals' foreign cash. We examine the real and financial response to this liquidity shock and find that firms did not increase capital expenditures, employment, R&D, or M&A, regardless of financial constraints. On the financial side, firms paid out only about one-third of the new liquidity to shareholders and retained half as cash. This high retention was not associated with poor governance. The high propensity to retain the liquidity shock as cash, even among well-governed firms with limited financial constraints, is difficult to reconcile with existing theory.

On January 1, 2018, the Tax Cuts and Jobs Act (TCJA) came into effect, changing several features of the corporate tax code. Among its key provisions was unlocking “trapped cash” that U.S. multinationals had accumulated in their foreign subsidiaries. Prior to the TCJA, the U.S. corporate tax code incentivized multinationals to defer repatriation of foreign income, leading to large balances of accumulated cash and financial securities (Foley et al., 2007; Faulkender et al., 2019). With the changes imposed by the TCJA, firms suddenly had access to additional internal capital at zero marginal cost, funds that previously had been extremely costly to use for U.S. investment or payouts. In

aggregate, the law increased liquidity of U.S. firms by providing access to \$1.7 trillion in foreign cash (Bureau of Economic Analysis, 2020).

In this paper, we evaluate how this sudden and significant increase in internal liquidity, brought about by the TCJA, affected real and financial decisions of U.S. multinationals. Although the tax law applied to all U.S. corporations, firms had differential exposure to this policy change based on their accumulated stock of liquid foreign profits. We use confidential data on cash and marketable security holdings at foreign subsidiaries of U.S. multinationals (hereafter, foreign cash) collected by the Bureau of Economic Analysis (BEA) to measure each

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firm's exposure to the TCJA-induced liquidity shock. We estimate firms' investment, labor, and merger and acquisition decisions in response to this liquidity increase using a difference-in-differences research design. In addition, we explore how firms adjusted their payouts, cash holdings, and leverage in response to the decreased cost of accessing their foreign cash holdings. Our results provide insight into how liquidity relates to firm decisions and whether the TCJA was successful in stimulating U.S. investment and hiring by unlocking firms' foreign cash. On the financial side, it reveals the degree to which firms actively adjust their cash holdings, shareholder payouts, and debt positions in response to internal liquidity shocks.

We find no evidence that firms responded to the liquidity shock by increasing investment in the U.S.: the increased liquidity led to no change in capital expenditure, wage expense, or R&D. Similarly, we find no effect on foreign real activity.¹ Theory predicts that real investment should increase if the firm is financially constrained or faces agency conflicts. However, we do not find that the most constrained firms responded differently than the least constrained. Similarly, the response does not vary with the strength of governance. This evidence is consistent with our sample of firms being neither materially financially constrained nor poorly governed. In addition, we do not find a differential response for firms in industries more amenable to voluntary profit shifting. Overall, our results indicate that a lack of liquidity did not constrain the real decisions of firms, and unlocking trapped cash likely did not stimulate investment and hiring as proponents of the policy had predicted.² Our findings also contrast with early evidence of a positive relationship between trapped cash and investment following the TCJA (Beyer et al., 2021; Olson, 2021; Markarian and Crawford, 2022). However, our study is unique in terms of identification strategy and data sources.³

We next ask whether increased internal liquidity led to increased acquisition activity. Similar to other investment decisions, increased liquidity could decrease the cost of capital (Myers, 1984) or exacerbate agency conflicts (Jensen, 1986), with each channel predicting increased M&A activity. Indeed, the merger literature has demonstrated that both firm-level and macro liquidity are significant drivers of merger activity (Harford, 1999, 2005). Specific to our setting, previous studies have shown that trapped cash was associated with increased foreign acquisitions prior to the TCJA, and that these acquisitions had lower announcement returns (Hanlon et al., 2015; Edwards et al., 2016). However, we find no evidence that firms responded to the liquidity shock by adjusting acquisition activity either in the U.S. or abroad. Further, we find a similar non-response for firms that have weak governance. While the findings of Harford et al. (2017) and Albertus et al. (2022) suggest an important role of agency conflicts in the firm's use of foreign cash, our results do not provide clear evidence of these conflicts.

¹ Consistent with our findings, Albertus et al. (2022) predicts that the response in foreign investment to the TCJA, while negative unconditionally, does not depend on the quantity of freed foreign cash.

² Tim Cook, CEO of Apple, was an outspoken critic of the pre-TCJA tax system because it discouraged repatriation: "If you earn money globally, you cannot bring it back into the United States unless you pay 35 percent plus your state tax. And you look at this and you go, 'This is kind of bizarre'. You want people to use this money in the United States to invest more. ... I think the [Trump] administration, you saw that they are really getting this and want to bring this back..." (CNBC, 2017). President Trump's economic advisors stated in the *New York Times* that freeing foreign cash would "attract more than \$2 trillion to these shores, raising billions for the Treasury while creating new jobs". (Forbes et al., 2017.)

³ Beyer et al. (2021) does not measure foreign cash but uses all U.S. multinationals as treated firms and domestic-only firms as controls. Olson (2021) only considers the two quarters following the reform and uses a measure of accumulated foreign profits to proxy for trapped cash. Markarian and Crawford (2022) considers the effect on consolidated capital expenditures and proxies for foreign cash using permanently reinvested earnings.

How did firms adjust their cash positions and shareholder payouts in response to the increase in internal liquidity? Traditional precautionary savings theory predicts that for a firm that is financially unconstrained—whether due to high cash holdings or low cost of external financing—the marginal value of liquidity approaches one: shareholders value a dollar of cash the same inside the firm or in their pockets (Riddick and Whited, 2009; Bolton et al., 2011). Add to this any cost of holding cash (e.g., agency costs or cost of carry) and shareholders prefer any additional liquidity to be paid out.

Indeed, existing empirical evidence suggests that unconstrained firms would respond to liquidity shocks by paying it out to shareholders rather than retaining it (Dharmapala, 2024). Notably, Almeida et al. (2004) finds that unconstrained firms do not retain cash inflows. Similarly, Dharmapala et al. (2011) finds that 92% of repatriations unlocked by the American Jobs Creation Act (AJCA) in 2005 were used to make shareholder payouts rather than being retained as cash. From this perspective, precautionary savings theory predicts that the cash unlocked by the TCJA should be paid out if these firms are financially unconstrained.

In contrast, we find that for each dollar of foreign cash that was freed by the TCJA, only about 30 cents was paid out to shareholders, primarily as share repurchases. This coincided with a 48 cent reduction in consolidated cash holdings. In addition, we do not find a significant response in firm leverage, suggesting that firms did not use this increased liquidity to pay down debt. Together, our findings reveal a high propensity of seemingly unconstrained firms to retain liquidity shocks as cash. The low payout and high cash retention we document is difficult to rationalize as a traditional precautionary savings motive for cash holdings.

To understand this result, we next consider potential frictions that may be driving the low payout and high retention of cash following the tax reform. First, firms may have held cash in their foreign subsidiaries for operational reasons, rather than due to a tax lockout effect. To test this possibility, we measure the liquidity shock as only the portion of foreign cash that is in excess of that required to support foreign operations. We find similar results using this alternative measure, suggesting that the small response is not due to the inclusion of foreign cash used for ongoing operations. Second, we consider whether managers are retaining the freed cash for the private benefits of control. We find no evidence that the financial response is due to agency conflicts: better governed firms reduce cash and increase payouts at the same level as poorly governed firms. Third, it is possible that adjusting the cash position and payout levels takes longer than two years, the observation window used in our benchmark sample. To test this, we extend our sample by two years through 2021. We do not find that adjustments to cash holdings, or elevated shareholder payouts, continue to occur more than two years following the reform. Taken together, the low payout and high cash retention in response to the liquidity increase does not appear to result from cash held for operations, agency conflicts, or a delay in adjustment.

Our evidence of high retention of new liquidity by financially unconstrained firms stands in tension with traditional precautionary savings theory, which suggests that unconstrained firms should pay out the new liquidity. However, our findings are consistent with what Graham and Leary (2018) describe as firms choosing to only "loosely" adhere to cash targets. As evidence of this loose adherence, Graham and Leary (2018) document that about half of the time-series variation in cash holdings is explained by contemporaneous cash flows and investment expenditures, consistent with only a partial adjustment in cash holdings in response to liquidity shocks. This parallels evidence of minimal active adjustment of debt in response to shocks to market leverage documented in Welch (2004). In the case of leverage adjustment, the muted response may be explained by fixed costs of adjustments (Fischer et al., 1989). However, in the case of cash holdings, this is an unlikely explanation, as paying out excess cash is not particularly costly.

One possible explanation that reconciles our findings with precautionary savings theory is that cash may act as a type of collateral for firm borrowing both before and after the reform. In addition to the usefulness of cash in satisfying short-term liquidity needs, Gao et al. (2021) show that firms may hold cash to lower the cost of borrowing by reducing the likelihood of—and increasing recovery in—default. This role for cash persists even after the cash was unlocked by the TCJA, potentially making it suboptimal to fully pay out these holdings.

Another possible explanation is that optimal cash holdings may be roughly indeterminate following the TCJA for firms in our sample because the marginal costs (i.e., costs of carry and agency) and benefits (i.e., precautionary savings) are both likely quite small for these firms.⁴ In other words, firms may have had strong tax incentives to accumulate cash prior to reform due to the repatriation tax, but little compelling reason to adjust cash downward following the reform. This weak incentive to adjust following reform could present as a loose cash target, but nevertheless be consistent with precautionary savings theory.

Our results provide new insight into firms' elevated cash holdings over the last several decades (Bates et al., 2009).⁵ Several papers—using data prior to the TCJA—have shown that tax incentives are an important driver of firms' cash holdings, and in particular foreign cash holdings, since the beginning of the century (Foley et al., 2007; Gu, 2017; Graham and Leary, 2018; Faulkender et al., 2019; Darmouni and Mota, 2024). We find that firms did reduce cash holdings in response to the removal of the repatriation tax, consistent with tax incentives contributing to firms' accumulation of cash. At the same time, the modest magnitude of the downward adjustment could be interpreted as evidence that factors beyond tax incentives are important to understanding patterns in cash holdings over the last two decades. From this perspective, our findings are consistent with Pinkowitz et al. (2016), who challenge a prominent view in the literature by asserting that repatriation taxes have only a limited ability to explain the high cash holdings of U.S. firms.

We contribute to a large literature seeking to understand the effect of liquidity on firm investment and financial policy (e.g., Fazzari et al., 1988; Almeida et al., 2004). The setting provided by the TCJA tax reform provides several advantages that help to overcome the empirical challenges inherent in answering these questions (Kaplan and Zingales, 1997; Erickson and Whited, 2000; Gomes, 2001; Altı, 2003). First, the TCJA liquidity shock is firm-specific, generating cross-sectional variation in the treatment intensity. This also allows for the inclusion of firm fixed effects in our regression specification, substantially controlling for unobserved firm heterogeneity. Second, our approach controls for average industry-year variation in the outcome variables resulting from other provisions of the TCJA—such as the reduction in the U.S. corporate income tax rate—allowing us to isolate the effect of liquidity changes. Third, the freeing of unrepatriated foreign earnings was mandatory, and therefore there are no endogenous selection effects driven by a firm's decision to participate in taxable repatriation. This contrasts with the AJCA, a previous voluntary repatriation program.

⁴ For example, the reduction in the corporate income tax rate made retaining cash within the firm less expensive post-TCJA, as the spread between the personal and corporate rates dropped significantly (Looney, 2017). In addition, firms likely faced a low cost of carry due to low interest rates during this period.

⁵ A large literature seeks to understand corporate liquidity and cash holdings decisions of firms (Opler et al., 1999; Riddick and Whited, 2009; Bolton et al., 2011; Denis, 2011; Almeida et al., 2014). Evidence suggests that precautionary savings motives (e.g., Han and Qiu, 2007; Gamba and Triantis, 2008; Riddick and Whited, 2009; Duchin et al., 2010; Almeida et al., 2011; Acharya et al., 2012; Cunha and Pollet, 2020), agency conflicts (Gao et al., 2013; Nikolov and Whited, 2014; Harford et al., 2017), interest rates (Azar et al., 2016; Gao et al., 2021), and an increase in intangible assets (Lyandres and Palazzo, 2016; Begenau and Palazzo, 2021; Falato et al., 2022) may all play a role in cash holdings decisions.

Fourth, we contribute to the literature by using novel data ideally suited to evaluating the effects of the TCJA on corporate activity. In particular, BEA's data on foreign cash holdings provide a sharp measure of the liquidity held in the foreign affiliates of U.S. multinationals.

A potential concern with our estimation approach is that our treatment variable—the quantity of foreign cash prior to the reform—may be correlated with the change in investment opportunities associated with the reform. Specifically, firms that had more foreign cash unlocked by the TCJA (e.g., due to higher past foreign profitability) may have also benefited more from a separate provision of the reform that eliminated U.S. taxes on foreign operations going forward.⁶ It is important to note that this potential omitted variable bias would tend to *overstate* the effect, of which we find none, mitigating this concern *ex post*. Nevertheless, we consider this potential bias by constructing a firm-specific proxy for the effect of the TCJA on investment opportunities. In particular, we estimate the wedge between U.S. and foreign tax rates for each firm's foreign operations: a higher value indicates that a firm may have a higher increase in post-TCJA investment opportunities than a firm with a lower wedge. We find that our results are unchanged with the inclusion of this control, providing evidence that our treatment variable is not acting as a proxy for post-reform investment opportunities.⁷ This is consistent with Garcia-Bernardo et al. (2022), who find that the TCJA did not significantly alter the geographical allocation of the foreign profits of U.S. multinationals. Further, we provide a dynamic analysis showing parallel trends in investment and payouts prior to treatment.

This paper relates to previous work that used a 2005 temporary repatriation tax holiday to determine how repatriated foreign cash was used by U.S. multinationals. The tax holiday, part of the AJCA, allowed firms to voluntarily repatriate funds, under certain restrictions, at a reduced tax rate. Dharmapala et al. (2011) finds that each dollar of increased repatriation was associated with up to a 92 cent increase in shareholder payouts, and found no effect on real investment even among constrained firms. In contrast, Faulkender and Petersen (2012) finds a small or insignificant increase in payouts while real investment increased among constrained firms. One of the primary empirical challenges of exploiting this 2005 program is that participation was voluntary, leading to an endogenous selection effect. These studies differ in their approach to addressing this selection problem, potentially leading to their contrasting results.⁸ Our environment avoids this concern altogether because participation was mandatory for all firms.⁹ Our findings contrast with these previous studies in that firms responded to the increased liquidity with only a modest increase in payouts and no increase in investment. In addition, neither study explores the effect on firms' cash holdings, where we find that firms choose to only partially

⁶ Note that our difference-in-differences specification already controls for average industry-year changes in investment opportunities, including those potentially brought about by the TCJA. The above concern only applies if the change in investment opportunities is correlated with our treatment variable and not controlled by firm fixed effects, industry-by-year fixed effects, or other controls.

⁷ A related potential concern is that firms with fewer foreign investment opportunities during the pre-period may accumulate more foreign cash, making the treatment intensity correlated with pre-TCJA investment opportunities. However, our identification strategy does not require the level of investment opportunities to be equal across firms as this heterogeneity will be removed by firm fixed effects (and other controls).

⁸ Faulkender and Petersen (2012) provide a detailed discussion of the differing estimation approaches, including Blouin and Krull (2009) who find that share repurchases increase for firms choosing to participate in repatriation.

⁹ The TCJA also resulted in a larger increase in liquidity than the voluntary repatriation holiday of 2005. Graham and Leary (2018) shows that firms only began to accumulate foreign earnings beginning around 2000 as average foreign tax rates—which have trended downward over the last few decades—dropped below the U.S. tax rate.

“undo” the effects of the liquidity increase by reducing cash holdings, with about half of the freed cash being retained by the firm.¹⁰

1. Data

To understand firms’ response to a liquidity shock provided by new access to their foreign cash, we require high quality non-consolidated data on firms’ domestic and foreign operations and financial positions. Unfortunately, such data are not publicly available. For example, data in Compustat’s geographic segment file are reported inconsistently across firms and are usually limited to information on sales. To address the shortcoming of publicly available data, one of the authors applied for and was granted access to the confidential income statements and balance sheets of U.S. multinationals’ domestic and foreign operations, maintained on secure government servers at the Bureau of Economic Analysis.

We construct our dataset from BEA’s annual and benchmark surveys of the Activities of U.S. Multinational Enterprises.¹¹ The surveys collect separate information on both the domestic and foreign operations of U.S. multinationals and are conducted pursuant to the International Investment and Trade in Services Survey Act (hereafter the Act). The Act stipulates that the “use of an individual company’s data for tax, investigative, or regulatory purposes is prohibited”. Willful noncompliance with the Act may result in imprisonment for up to one year. For these reasons, in addition to their monitoring of corporate events and a system of internal data integrity checks, BEA believes the surveys accurately capture virtually complete data on the universe of U.S. direct investment abroad.

We limit the sample to majority-owned affiliates, which are commonly referred to as “subsidiaries”, the term used in this paper. In addition, we omit firms in the financial services (NAICS 5200–5299), utility (NAICS 2200–2299), and public administration (NAICS 9200–9299) industries. We also require that firms have at least one subsidiary in a country with a corporate income tax rate below the U.S. rate prior to 2018. We merge these data with Compustat, SDC, and FactSet to incorporate information on stock repurchases, mergers and acquisitions, and institutional equity ownership, respectively. We start the sample in 2010 to include several years prior to the TCJA while avoiding the turbulence of the financial crisis. In the baseline analysis we end the sample in 2019, which is the last year for which BEA’s data are available. The baseline sample consists of 3146 firm-year observations, corresponding to 489 unique firms. For certain analyses that rely on consolidated (Compustat) outcomes, we extend the sample through 2021.

In , we present descriptive statistics for the consolidated firms (Panel A), their domestic operations (Panel B), and their foreign operations (Panel C). We scale dollar-denominated stock variables (e.g., cash) by consolidated book value of assets and flow variables (e.g., dividends) by lagged consolidated book value of assets. Throughout the paper, financial variables are recorded in millions of 2018 U.S. dollars. In addition, we winsorize each variable at its 1st and 99th percentiles to mitigate the influence of outliers. [Appendix A](#) contains detailed definitions for all variables.

Panel A indicates that the ratio of consolidated cash to consolidated assets (*Cash*) averages 0.151. Sample firms’ foreign subsidiaries tend to face relatively low tax rates. Specifically, the U.S. statutory tax rate in 2017 (35%) less the sales weighted average foreign statutory tax rate in 2017 (*TaxGap*) averages 10.5%. Panel B shows that domestic

Table 1

Descriptive statistics. The table presents descriptive statistics for the baseline sample, which starts in 2010 and ends in 2019. Panels A, B, and C correspond to sample firms’ consolidated operations, U.S. operations, and foreign operations, respectively. In each case, the sample consists of 3146 firm-year observations. [Appendix A](#) contains variable definitions.

Panel A: Consolidated operations		
	Mean	Std. dev.
<i>Cash</i>	0.151	0.142
<i>Dividends</i> (to shareholders)	0.018	0.022
<i>Repurchases</i>	0.037	0.054
<i>Payouts</i>	0.055	0.065
<i>Leverage</i>	0.264	0.175
<i>TCJA</i>	0.228	0.420
<i>TaxGap</i>	0.105	0.040
<i>Age</i>	3.463	0.636
<i>Market-to-book</i>	2.021	1.128
<i>Profitability</i>	0.053	0.074
<i>HP</i> (Hadlock Pierce)	0.425	0.298
<i>Ownership</i>	0.298	0.079
<i>EffectiveTaxRate</i>	0.233	0.201
<i>Intangibility</i>	0.622	0.265
Panel B: Domestic operations		
	Mean	Std. dev.
<i>U.S. investment</i>	0.026	0.028
<i>U.S. employment</i>	0.132	0.099
<i>U.S. R&D</i>	0.026	0.039
<i>U.S. M&A</i>	0.005	0.029
Panel C: Foreign operations		
	Mean	Std. dev.
<i>Foreign investment</i>	0.012	0.015
<i>Foreign employment</i>	0.054	0.056
<i>Foreign R&D</i>	0.006	0.011
<i>Foreign M&A</i>	0.001	0.008
<i>Repatriations</i>	0.007	0.025
<i>Foreign cash</i> (treatment)	0.050	0.056

capital expenditures (*U.S. investment*) is roughly one fifth as large as domestic employment compensation (*U.S. employment*). Finally, Panel C indicates that foreign cash makes up about one-third of total cash: the ratio of foreign cash to consolidated assets in 2017 (*Foreign cash*) is 0.050, compared to a consolidated cash ratio of 0.151. Repatriations from a firm’s foreign subsidiaries to its U.S. parent, scaled by lagged consolidated assets, (*Repatriations*) averages 0.7% annually.

2. Empirical design

We study a provision of the TCJA that eliminated the repatriation tax on U.S. multinational firms’ accumulated foreign profits. With this change, firms suddenly had access to additional internal capital at zero marginal cost, funds that previously had been extremely costly to use for U.S. investment or payouts.¹²

¹² Although access to the additional capital became costless *on the margin* following the reform, the law did impose a sunk cost on firms. Specifically, prior to the TCJA, firms paid U.S. taxes on their foreign earnings upon repatriation. Repatriation taxes could be deferred indefinitely by keeping the profits in their foreign subsidiaries and not using the funds for U.S. investment or shareholder payouts. The TCJA imposed a one-time, mandatory “transition tax” on all previously unrepatriated earnings, automatically freeing all foreign cash. The legislation provided a tax holiday on this deferred tax liability in that the transition tax rate on foreign financial assets, at 15.5% before foreign tax credits, was significantly lower than the (voluntary) repatriation tax rate prior to 2018. The legislation also removed repatriation taxes going forward. See [Dobridge et al. \(2023\)](#) and [Dharmapala \(2024\)](#) for details on the provisions of the TCJA.

¹⁰ Our results on payout policy also relate to [Chetty and Saez \(2005\)](#) and [Yagan \(2015\)](#), who find that the dividend tax rate is negatively associated with dividend payments. Similarly, we find that the TCJA, which reduced the marginal tax cost of distributions to shareholders, led to an increase in payouts.

¹¹ These data are collected for the purpose of producing publicly available aggregate statistics on the activities of multinational enterprises.

The TCJA, as legislation, emerged rapidly and was passed and implemented in short order. The earliest draft of the text was introduced in the U.S. House of Representatives on November 2, 2017. The final bill was signed into law on December 22, 2017, and became effective less than two months after it was introduced on January 1, 2018. Earlier in 2017 there was discussion of tax reform, but whether it would happen, when, and what would be included in a possible law (particularly, whether it would unlock firms' foreign cash) were unknown.¹³ Indeed, the details of the legislation were being negotiated even in the final hours before the vote in the Senate, with holdouts demanding changes that had to be handwritten in the margins of the bill to secure a slim 51-vote passage (Jansen, 2017; Pramuk, 2017). As a result, in conjunction with the fixed effects and control variables described below, we believe the TCJA generated a plausibly exogenous shock to firms' internal liquidity.

To estimate the response in real and financial variables following the TCJA liquidity shock, we implement a difference-in-differences research design. Our baseline specification is

$$Y_{i,t} = \beta \times TCJA_i \times ForeignCash_i + \gamma' X_{i,t-1} + \delta_{j,t} + \alpha_i + \varepsilon_{i,t}, \quad (1)$$

where $Y_{i,t}$ is the outcome variable of interest, such as investment, payouts, and cash holdings, for firm i in year t . The treatment intensity ($ForeignCash_i$) is the firm's foreign cash holdings scaled by total firm assets in 2017, our proxy for the amount of liquidity unlocked by the TCJA.¹⁴ The indicator $TCJA_i$ equals one after the TCJA was implemented (i.e., in 2018 and later), and is zero in other years.

Critically, our baseline specification includes firm fixed effects (α_i). These remove unobserved firm-level heterogeneity that may be correlated with foreign cash accumulation and drive investment decisions. In addition, industry-by-year fixed effects ($\delta_{j,t}$), where industry is categorized by four-digit NAICS codes, help control for potential industry-specific responses to the implementation of the TCJA.

Since one of our primary goals is to identify the effect of liquidity increases on real investment, we must ensure that our treatment intensity measure does not act as a proxy for unobserved investment opportunities. Firm and industry-by-year fixed effects control for both (1) time-invariant cross-sectional differences in investment opportunities, and (2) industry-average changes in investment opportunities over time, including those brought about by the TCJA. These controls largely address several concerns, including that the TCJA included a tax rate cut on U.S. income, potentially improving after-tax investment returns for all firms. Still, there may be a firm-specific change in investment opportunities brought about by the TCJA. If (and only if) this firm-specific change is correlated with pre-TCJA foreign cash accumulation, and is omitted from the specification, then this could bias our estimates.

It is worth noting that this omitted variable bias would generally overstate the response of investment to the liquidity increase because foreign cash is likely to be positively correlated with the firm-specific increase in foreign investment opportunities. For example, a firm that generates most of its income in a low-tax country would benefit more from the removal of the repatriation tax provided by the TCJA.¹⁵

¹³ Several studies examine the effect of anticipation, for example, De Simone et al. (2019), Borochin et al. (2021), Diercks et al. (2021), Gallemore et al. (2021), and Albertus et al. (2022).

¹⁴ *ForeignCash* is correlated with the 15.5% transition tax on foreign profits held as cash also imposed by the TCJA. We estimate that this is reduced to 4.2% on average after allowable foreign tax credits. However, the transition tax could be paid in instalments over eight years, minimizing this levy as a cash flow shock in itself. Using an "adjusted" cash flow shock that accounts for the transition tax would yield virtually identical inferences. Specifically, it would reduce the magnitude of *ForeignCash* by $\frac{4.2\%}{8} \approx 0.5\%$, with a similarly small effect on the magnitude of our point estimates.

¹⁵ Note that foreign cash holdings may also be high if foreign investment opportunities are low, leading to a bias in the other direction. However, unless the investment opportunities change with the passage of the TCJA, this pattern would be removed by the firm fixed effects and hence not impact our estimates.

Nevertheless, to address this concern, we construct a control for the firm-specific change in investment opportunities generated by the TCJA based on the source of each firm's foreign income. In particular, we define the firm's *TaxGap* as the U.S. statutory corporate tax rate minus the average statutory tax rate faced by the firm's subsidiaries, where the average is weighted by subsidiary sales, constructed using 2017 data. Because the measure is intended to capture the change in post-reform investment opportunities, the tax gap control is set to zero prior to 2018. That is, we use $TCJA \times TaxGap$ in the regression specifications.¹⁶ The tax gap measures the reduction in the repatriation tax rate brought about by the TCJA: a higher value corresponds to a larger increase in after-tax returns on future foreign investment. Thus, the tax gap measure is likely to capture firm-specific changes in investment opportunities that may be associated with the TCJA.¹⁷ We use statutory tax rates in the calculation, as effective tax rates are likely influenced by firms' choices (Dharmapala, 2014).

Our baseline regressions (Tables 3, 4, and 11), and many additional analyses, include this $TCJA \times TaxGap$ control, as well as a control for firm age. The latter is not directly chosen by the firm, at least on an ongoing basis, and is therefore arguably exogenous to the outcomes we study. Partly for this reason, it commonly appears in the literature (e.g., Giroud, 2013). We define age as the logarithm of 1 plus the number of years since the firm first appears in the Compustat data.

In the regressions with real outcomes, we add additional controls that often appear in the empirical literature analysing real economic activity. Specifically, we include controls for the lagged market-to-book ratio (Dharmapala et al., 2011; Almeida and Campello, 2007; Graham et al., 2017) and lagged firm profitability (Shin and Stulz, 1998; Badertscher et al., 2013; Shroff et al., 2014).

We are also sympathetic to the view that controls may be correlated with both our treatment and $Y_{i,t}$, leading to biased estimates of the treatment effect (Gormley and Matsa, 2011; Roberts and Whited, 2013). Accordingly, we re-estimate our main specifications omitting all controls. We discuss the results, which are qualitatively and quantitatively similar to the baseline, in Section 3.

Our treatment variable, *ForeignCash*, uses unconsolidated financial data from BEA to measure the cash and marketable securities held in the foreign subsidiaries of U.S. firms in 2017. This measure provides advantages in our setting relative to alternative measures. One measure commonly used in the literature is the foreign income that the firm chooses to designate, for accounting purposes, as "permanently reinvested earnings" (PRE), which may be held as financial assets or invested in real assets.¹⁸ PRE includes substantial real investments that will not increase firm liquidity following the TCJA. In addition, foreign profits that are not designated as PRE can still be held in foreign subsidiaries, as cash or real assets. In contrast, our measure ensures that we are capturing cash and financial assets, rather than illiquid real investments, that truly enhance a firm's liquidity position, regardless of the firm's chosen PRE accounting designation.¹⁹

¹⁶ In addition, the tax gap measure is set to zero if the difference takes a negative value because the firm likely faces no repatriation tax on foreign income.

¹⁷ The TCJA also lowered the U.S. corporate income tax rate, likely leading to increased investment opportunities. However, unlike the tax rate on foreign profits, all firms faced the same reduction in the statutory tax rate on domestic income. In our specification, industry-by-year fixed effects control for any change that affected all firms within an industry.

¹⁸ See, for example, Faulkender and Petersen (2012) and Markarian and Crawford (2022).

¹⁹ See Laplante and Nesbitt (2017) for a detailed discussion of the differences between permanently reinvested earnings, foreign cash, and trapped cash, and Blouin et al. (2022) for a description of where U.S. multinationals hold PRE.

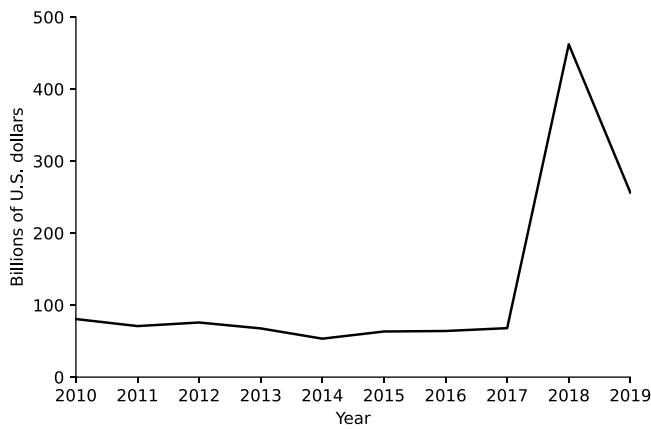


Fig. 1. Aggregate repatriations. This figure plots annual aggregate repatriations of U.S. multinationals' foreign subsidiaries to their U.S. parents. Values are in billions of 2018 U.S. dollars.

For most outcome variables, we also explore the dynamics of the treatment effect by estimating the response using year-by-year indicators in lieu of the single $TCJA_t$ indicator:

$$Y_{i,t} = \sum_{s \in \mathcal{T}} (\beta_s \times \mathbb{1}_{[t=s]} \times ForeignCash_i) + \gamma' X_{i,t-1} + \delta_{j,t} + \alpha_i + \varepsilon_{i,t}, \quad (2)$$

where $\mathcal{T} = \{2010 - 2014, 2015, 2016, 2018, 2019\}$, that is, a five-year pre-period plus the remaining years in our sample excluding 2017 (the reference year).

3. Results

In this section, we first estimate the effect of the TCJA-induced increase in internal liquidity on firm's real investment decisions, including capital expenditures, employment, research and development (R&D), and mergers and acquisitions (M&A). We consider whether financial constraints or agency conflicts play a role in this response. Next, we explore the response in the financial positions of firms following the unlocking of their foreign cash, including changes in payouts, cash holdings, and leverage.

Before proceeding with our main empirical tests, we first explore whether our treatment variable—a firm's foreign cash ratio—predicts subsidiary-to-parent repatriations. Following the TCJA, the distinction between foreign and domestic cash was effectively nullified for U.S. tax purposes, meaning that a firm could pay dividends from its foreign subsidiaries to its U.S. parent (i.e., a repatriation) without tax consequences in the U.S. If firms were holding cash in their foreign subsidiaries due to the tax lockout created by repatriation taxes prior to 2018, we might expect to see foreign cash holdings partially reallocated to U.S. parents.

Table 2 shows the results of our baseline regression using Eq. (1) with subsidiary-to-parent repatriations as the outcome variable. Column (1) shows that 13 cents of each dollar of foreign cash was repatriated annually in the two years following the reform, meaning roughly one quarter of the foreign cash was repatriated. The year-by-year estimates, using Eq. (2), are reported in Column (2), indicating that considerable repatriation occurs each year post-TCJA. The increase in repatriations following reform can also clearly be seen in the unconditional dollar amount of repatriations shown in Fig. 1.²⁰

The reallocation of cash from foreign subsidiaries to U.S. parents provides suggestive evidence that taxes played a role in firms' decision to accumulate foreign cash. We next explore how the freed foreign cash affected the real and financial decisions of firms.

Table 2

Repatriations. This table presents estimates of the association between firms' foreign cash and their repatriations following the TCJA. The unit of observation is the firm-year. *Repatriations* is the dividends from a firm's foreign operations to its U.S. parent scaled by the firm's lagged consolidated assets. *TCJA* is an indicator equalling 1 in 2018 and later. *ForeignCash* is the ratio of a firm's foreign cash in 2017 to the firm's consolidated assets in 2017. $\mathbb{1}_{year \leq 2014}$ is an indicator equalling one in 2014 and before. $\mathbb{1}_{year=2015}$ is an indicator equalling one in 2015. $\mathbb{1}_{year=2016}$ through $\mathbb{1}_{year=2019}$ are defined analogously. Controls are *Age* and *TCJA* $\times$ *TaxGap*, which are defined in Appendix A. Fixed effects are for firm and industry-by-year. Standard errors are clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Repatriations	
	(1)	(2)
<i>TCJA</i> $\times$ <i>ForeignCash</i>	0.125** (0.051)	
$\mathbb{1}_{year \leq 2014} \times ForeignCash$		−0.005 (0.033)
$\mathbb{1}_{year=2015} \times ForeignCash$		−0.037 (0.040)
$\mathbb{1}_{year=2016} \times ForeignCash$		−0.013 (0.023)
$\mathbb{1}_{year=2018} \times ForeignCash$		0.110* (0.064)
$\mathbb{1}_{year=2019} \times ForeignCash$		0.127* (0.075)
<i>Age</i>	−0.011 (0.008)	−0.011 (0.008)
<i>TCJA</i> $\times$ <i>TaxGap</i>	0.009 (0.050)	0.009 (0.051)
Fixed effects	Yes	Yes
R ²	0.558	0.558
Observations	3146	3146

3.1. Real economic activity

Did removing the tax lockout on foreign cash stimulate investment and employment by U.S. multinationals? We assess the effect of the increased liquidity on firms' real activity by estimating the difference-in-differences specification in Eq. (1). Table 3 reports the results for domestic (U.S.) investment and employment. We define investment as capital expenditures and employment as wage expense, each scaled by lagged assets.

Columns (1) and (3) report the results for the average treatment effect in the two-year post period. Across both measures of U.S. real activity, we find no significant response by firms to the increased access to foreign cash. Contrary to the hopes of some TCJA proponents, these results indicate that the reform likely did not have a discernible effect on the real decisions of firms in our sample. The findings suggest that these firms are not meaningfully constrained financially and thus the increased liquidity did not materially decrease the cost of capital.

We explore the dynamics of the treatment effect by estimating the response using year-by-year indicators as in Eq. (2). The results are reported in Columns (2) and (4) of Table 3; the year-by-year coefficients, along with 95% confidence intervals, are plotted in Panels A and B of Fig. 2. While the difference is significant at the 10% level in 2015 for U.S. employment, there does not appear to be a pre-trend. This is reassuring that U.S. real activity in the years leading up to the reform is not significantly related to foreign cash holdings immediately prior to the reform.

We find that the coefficient estimates on *TCJA* $\times$ *Tax Gap*—the proxy for the firm-specific change in foreign investment opportunities induced by the TCJA—are small and insignificant in all cases. This suggests that the change in tax treatment of foreign earnings did not heterogeneously affect firm investment in a significant way. These results are consistent with Garcia-Bernardo et al. (2022), who find

²⁰ Blouin et al. (2023) explores the accounting frictions related to PRE in the repatriation decision following the passage of the TCJA.

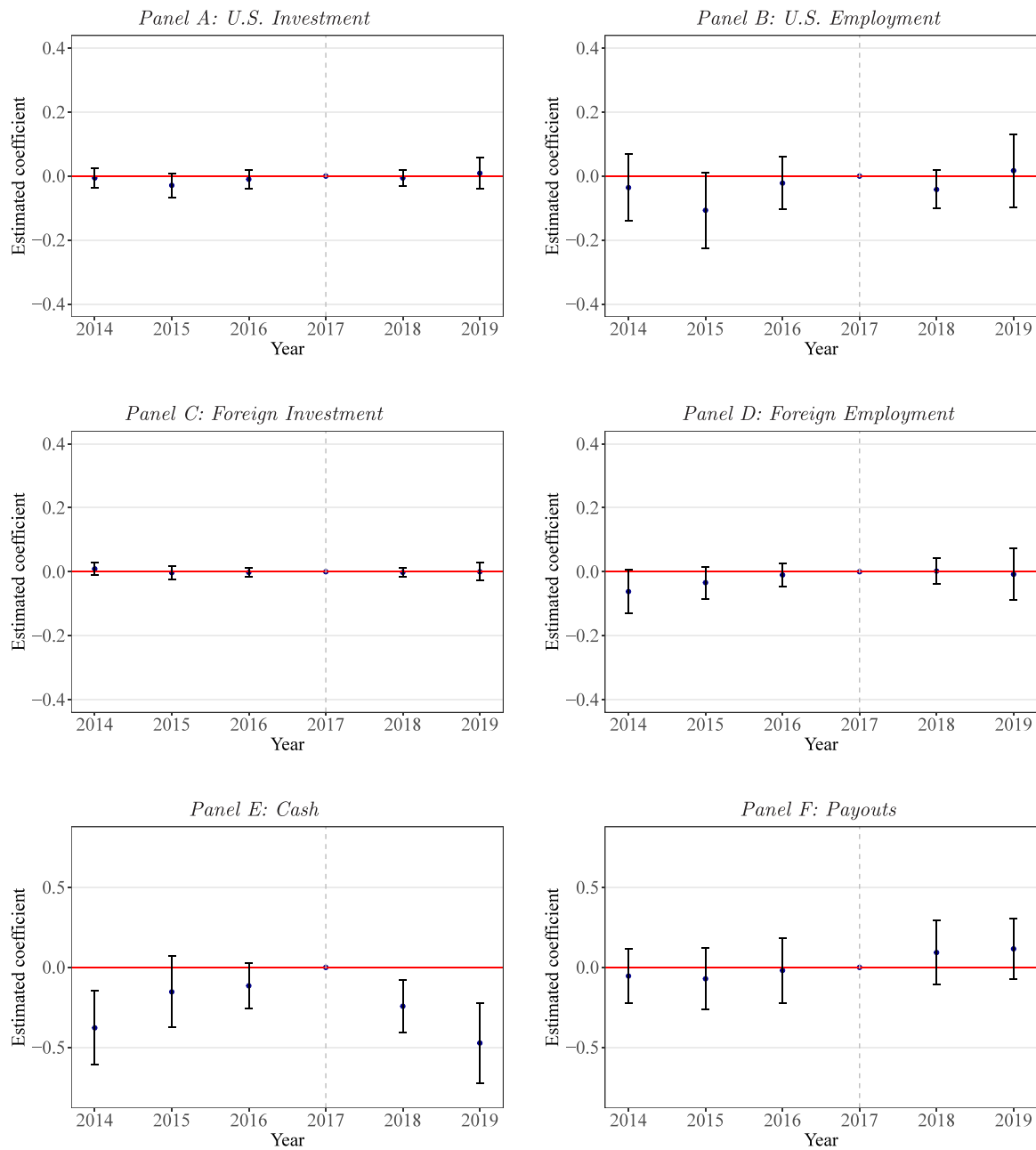


Fig. 2. Dynamic estimates. This figure plots the estimated coefficients, β_s , from Eq. (2). The dependent variable is U.S. investment (Panel A), U.S. employment (Panel B), foreign investment (Panel C), foreign employment (Panel D), cash holdings (Panel E), and total payout (Panel F). The vertical lines denote 95% confidence intervals. The coefficient corresponding to the years 2010–2014 in Eq. (2) is denoted by 2014 in each panel. Appendix A contains variable definitions.

that the TCJA did not significantly alter the geographic allocation of the foreign profits of U.S. multinationals. Regardless, the results are very similar with or without the inclusion of the control for *TaxGap*.²¹ Overall, this suggests that our results are not driven by heterogeneity in firm-specific changes in investment opportunities.

The tests are repeated using foreign real activity for the outcome variables in Table 4. The implications of increased U.S. parent access to foreign cash is unclear, as foreign investment could be made with this foreign cash prior to 2018 without incurring repatriation taxes. Albertus et al. (2022) finds that the worldwide tax system may have led to increased foreign investment prior to the TCJA—a prediction supported by Samuel (2022) who finds a post-TCJA decline in

foreign investment—but that this post-reform decline does not depend on the *quantity* of accumulated foreign cash. Consistent with this, and with our results for U.S. real activity, Table 4 shows no significant response in foreign real activity to the quantity of foreign cash. The year-by-year coefficients, along with 95% confidence intervals, are plotted in Panels C and D of Fig. 2.

We consider the robustness of our results by repeating the estimation without control variables, as well as using research and development (R&D) as an alternative measure of real economic activity. The results are shown in Table 5. In all cases, the results are similar to the findings in our baseline estimation. We also augment the baseline specification with the firm's effective tax rate, which we calculate as the ratio of its taxes paid to its pretax income. As seen in the Online Appendix Table OA.1, the estimates are very similar.

²¹ The results without controls are provided in Table 5.

Table 3

U.S. real activity. This table presents estimates of the association between firms' foreign cash and their U.S. real economic activity. *U.S. investment* is the ratio of U.S. capital expenditures to lagged consolidated assets. *U.S. employment* is the ratio of U.S. employment compensation to lagged consolidated assets. *TCJA* is an indicator equalling 1 in 2018 and later. *ForeignCash* is the ratio of a firm's foreign cash in 2017 to the firm's consolidated assets in 2017. $\mathbb{I}_{year \leq 2014}$ is an indicator equalling one in 2014 and before. $\mathbb{I}_{year=2015}$ is an indicator equalling one in 2015. $\mathbb{I}_{year=2016}$ through $\mathbb{I}_{year=2019}$ are defined analogously. Controls are *Age*, *TCJA* $\times$ *TaxGap*, *Market-to-book*, and *Profitability*, which are defined in Appendix A. Fixed effects are for firm and industry-year. Standard errors are clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>U.S. investment</i>		<i>U.S. employment</i>	
	(1)	(2)	(3)	(4)
<i>TCJA</i> $\times$ <i>ForeignCash</i>	0.007 (0.015)		0.012 (0.041)	
$\mathbb{I}_{year \leq 2014} \times \text{ForeignCash}$		−0.006 (0.015)	−0.036 (0.052)	
$\mathbb{I}_{year=2015} \times \text{ForeignCash}$		−0.029 (0.019)	−0.107* (0.059)	
$\mathbb{I}_{year=2016} \times \text{ForeignCash}$		−0.010 (0.015)	−0.022 (0.041)	
$\mathbb{I}_{year=2018} \times \text{ForeignCash}$		−0.006 (0.013)	−0.042 (0.030)	
$\mathbb{I}_{year=2019} \times \text{ForeignCash}$		0.009 (0.024)	0.017 (0.057)	
<i>Age</i>	−0.009* (0.005)	−0.009* (0.005)	−0.015 (0.016)	−0.015 (0.016)
<i>TCJA</i> $\times$ <i>TaxGap</i>	0.016 (0.026)	0.016 (0.026)	0.008 (0.061)	0.007 (0.062)
<i>Market-to-book</i>	0.004*** (0.001)	0.004*** (0.001)	0.013*** (0.003)	0.013*** (0.003)
<i>Profitability</i>	0.020** (0.008)	0.020** (0.008)	0.009 (0.021)	0.008 (0.022)
Fixed effects	Yes	Yes	Yes	Yes
R ²	0.858	0.858	0.912	0.913
Observations	3146	3146	3146	3146

3.1.1. Financial constraints, agency conflicts, and profit shifting

We find that firms do not respond to increased access to liquidity by increasing capital expenditures, employment, or R&D. However, basic financial theory predicts that investment would increase if the firm is financially constrained (Stein, 1997; Giroud and Mueller, 2015), whereby an increase in internal capital would lower the cost of capital. In contrast, investment would not increase for unconstrained firms. We explore this possibility by testing whether the investment response depends on financial constraints.

Financial constraints are notoriously difficult to measure (Farre-Mensa and Ljungqvist, 2016). With this caveat, we rely on the measure of Hadlock and Pierce (2010) based on firm size and age. We augment the baseline regression in Eq. (1) to include the interaction with financial constraints, specifically:

$$Y_{i,t} = \beta_0 \times TCJA_t \times ForeignCash_i \times HP_i + \beta_1 \times TCJA_t \times ForeignCash_i + \beta_2 \times TCJA_t \times HP_i + \gamma' X_{i,t} + \delta_{j,t} + \alpha_i + \varepsilon_{i,t} \quad (3)$$

HP is the Hadlock and Pierce (2010) measure of financial constraints. A positive estimate for β_0 would indicate that more financially constrained firms increase investment relative to unconstrained firms in response to the freeing of foreign cash.

Table 6 reports the results from estimating Eq. (3). For all outcome variables, both in the U.S. and overseas, the estimated coefficient on the interaction is insignificant. Overall, we find no evidence that financial constraints are a significant predictor of investment response to the TCJA liquidity shock. These findings further support our previous results indicating that U.S. multinationals are not significantly

Table 4

Foreign real activity. This table presents estimates of the association between firms' foreign cash and their foreign real economic activity. *Foreign investment* is the ratio of foreign capital expenditures to lagged consolidated assets. *Foreign employment* is the ratio of foreign employment compensation to lagged consolidated assets. *TCJA* is an indicator equalling 1 in 2018 and later. *ForeignCash* is the ratio of a firm's foreign cash in 2017 to the firm's consolidated assets in 2017. $\mathbb{I}_{year \leq 2014}$ is an indicator equalling one in 2014 and before. $\mathbb{I}_{year=2015}$ is an indicator equalling one in 2015. $\mathbb{I}_{year=2016}$ through $\mathbb{I}_{year=2019}$ are defined analogously. Controls are *Age*, *TCJA* $\times$ *TaxGap*, *Market-to-book*, and *Profitability*, which are defined in Appendix A. Fixed effects are for firm and industry-year. Standard errors are clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>Foreign investment</i>		<i>Foreign employment</i>	
	(1)	(2)	(3)	(4)
<i>TCJA</i> $\times$ <i>ForeignCash</i>	−0.005 (0.007)		0.033 (0.028)	
$\mathbb{I}_{year \leq 2014} \times \text{ForeignCash}$		0.009 (0.010)	−0.062* (0.034)	
$\mathbb{I}_{year=2015} \times \text{ForeignCash}$		−0.003 (0.010)	−0.034 (0.025)	
$\mathbb{I}_{year=2016} \times \text{ForeignCash}$		−0.002 (0.007)	−0.010 (0.018)	
$\mathbb{I}_{year=2018} \times \text{ForeignCash}$		−0.002 (0.007)	0.002 (0.020)	
$\mathbb{I}_{year=2019} \times \text{ForeignCash}$		−0.000 (0.014)	−0.008 (0.040)	
<i>Age</i>	−0.002 (0.002)	−0.002 (0.002)	−0.015 (0.010)	−0.015 (0.010)
<i>TCJA</i> $\times$ <i>TaxGap</i>	−0.001 (0.010)	−0.001 (0.010)	0.014 (0.026)	0.012 (0.026)
<i>Market-to-book</i>	0.002*** (0.000)	0.002*** (0.000)	0.003** (0.001)	0.003** (0.001)
<i>Profitability</i>	0.007 (0.005)	0.007 (0.005)	−0.001 (0.011)	−0.001 (0.012)
Fixed effects	Yes	Yes	Yes	Yes
R ²	0.829	0.829	0.933	0.933
Observations	3146	3146	3146	3146

constrained. This is consistent with the findings of Dharmapala et al. (2011) for U.S. multinationals, and with Farre-Mensa et al. (2022) for the universe of Compustat firms.

In addition to financial constraints, misaligned incentives between managers and shareholders may lead to an increase in real activity in response to the liquidity spike even if the firm is not financially constrained. We consider this possibility by repeating the estimation in Eq. (3) using a proxy for the strength of corporate governance in the place of financial constraints. We proxy for corporate governance using the fraction of equity ownership of the top 5 institutional investors, as defined by Ferreira and Matos (2008). The results are reported in Table 7. For U.S. investment, U.S. employment, and foreign employment, the interaction term is insignificant, suggesting that agency conflicts are not an important driver of real activity in response to increased access to cash. For foreign investment, an outcome with no clear prediction, the interaction term is statistically significant at the 10% level, but is economically small.²²

We next consider the possibility that the location of profits is (partially) chosen by firms. In particular, firms may use tax strategies to move profits to low tax jurisdictions (Hines and Rice, 1994). The accumulation of foreign cash may be the result of aggressive profit shifting or may simply be the result of organic overseas operations. We may expect the latter group—firms that did not voluntarily move

²² Increasing ownership by one standard deviation increases foreign investment by one cent for each dollar of foreign cash.

Table 5

Robustness of real effects. This table presents robustness tests of the results on real economic activity. *U.S. investment* is the ratio of U.S. capital expenditures to lagged consolidated assets. *U.S. employment* is the ratio of U.S. employment compensation to lagged consolidated assets. *Foreign investment* is the ratio of foreign capital expenditures to lagged consolidated assets. *Foreign employment* is the ratio of foreign employment compensation to lagged consolidated assets. *U.S. R&D* is the ratio of U.S. research and development expenditures to lagged consolidated assets. *Foreign R&D* is the ratio of aggregate foreign research and development expenditures to lagged consolidated assets. $\mathbb{I}_{year \leq 2014}$ is an indicator equalling one in 2014 and before. $\mathbb{I}_{year=2015}$ is an indicator equalling one in 2015. $\mathbb{I}_{year=2016}$ through $\mathbb{I}_{year=2019}$ are defined analogously. *ForeignCash* is the ratio of a firm's foreign cash in 2017 to the firm's consolidated assets in 2017. Controls are *Age*, $TCJA \times TaxGap$, *Market-to-book*, and *Profitability*, which are defined in Appendix A. Fixed effects are for firm and industry-year. Standard errors are clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>U.S. inv.</i> (1)	<i>U.S. emp.</i> (2)	<i>For. inv.</i> (3)	<i>For. emp.</i> (4)	<i>U.S. R&D</i> (5)	<i>For. R&D</i> (6)
$\mathbb{I}_{year \leq 2014} \times ForeignCash$	−0.002 (0.014)	−0.022 (0.056)	0.011 (0.010)	−0.059* (0.034)	−0.001 (0.027)	0.006 (0.012)
$\mathbb{I}_{year=2015} \times ForeignCash$	−0.030 (0.018)	−0.102* (0.061)	−0.003 (0.010)	−0.034 (0.025)	−0.030 (0.024)	−0.002 (0.006)
$\mathbb{I}_{year=2016} \times ForeignCash$	−0.009 (0.016)	−0.022 (0.044)	−0.002 (0.007)	−0.011 (0.017)	0.007 (0.021)	0.003 (0.005)
$\mathbb{I}_{year=2018} \times ForeignCash$	0.007 (0.013)	0.002 (0.031)	0.003 (0.007)	0.014 (0.019)	−0.013 (0.019)	0.000 (0.004)
$\mathbb{I}_{year=2019} \times ForeignCash$	0.035 (0.022)	−0.096* (0.054)	0.010 (0.014)	0.013 (0.039)	−0.028 (0.028)	−0.004 (0.008)
Controls	No	No	No	No	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.852	0.909	0.826	0.932	0.931	0.854
Observations	3146	3146	3146	3146	3146	3146

Table 6

Financial constraints and real activity. This table presents results on the role of financial constraints in mediating the association between firms' foreign cash and their real economic activity following the TCJA. *U.S. investment* is the ratio of foreign capital expenditures to lagged consolidated assets. *U.S. employment* is the ratio of foreign employment compensation to lagged consolidated assets. *Foreign investment* is the ratio of foreign capital expenditures to lagged consolidated assets. *Foreign employment* is the ratio of foreign employment compensation to lagged consolidated assets. *TCJA* is an indicator equalling 1 in 2018 and later. *ForeignCash* is the ratio of a firm's foreign cash in 2017 to the firm's consolidated assets in 2017. *HP* is the 2017 value of the financial constraint measure calculated following Hadlock and Pierce (2010). Controls are *Age*, $TCJA \times TaxGap$, *Market-to-book*, and *Profitability*, which are defined in Appendix A. Fixed effects are for firm and industry-year. Standard errors are clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>U.S. inv.</i> (1)	<i>U.S. emp.</i> (2)	<i>For. inv.</i> (3)	<i>For. emp.</i> (4)
$TCJA \times ForeignCash \times HP$	0.026 (0.043)	0.071 (0.160)	0.030 (0.022)	−0.028 (0.084)
$TCJA \times ForeignCash$	−0.007 (0.026)	−0.034 (0.100)	−0.022 (0.014)	0.046 (0.054)
$TCJA \times HP$	−0.003 (0.004)	0.021* (0.012)	−0.001 (0.002)	0.010** (0.005)
Controls	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes
R ²	0.858	0.913	0.829	0.933
Observations	3146	3146	3146	3146

their profits overseas—to respond more strongly to the freeing of their foreign cash.

To test this, we run the following regression:

$$Y_{i,t} = \beta_1 \times TCJA_i \times ForeignCash_i \times Intangibility_j + \beta_2 \times TCJA_i \times ForeignCash_i + \beta_3 \times TCJA_i \times Intangibility_j + \beta_4 \times Intangibility_j + \gamma' X_{i,t-1} + \delta_t + \alpha_i + \epsilon_{i,t} \quad (4)$$

We proxy for profit-shifting by using asset intangibility as a measure of the firm's ability to choose the location of its profits (e.g., Grubert, 2003; Albertus et al., 2019). Asset intangibility is measured in the year prior to the TCJA as the industry median ratio of intangible capital, as constructed by Peters and Taylor (2017), to assets. We expect β_1 to

Table 7

Agency conflicts and real activity. This table presents results on the role of agency conflicts in mediating the association between firms' foreign cash and their real economic activity following the TCJA. *U.S. investment* is the ratio of foreign capital expenditures to lagged consolidated assets. *U.S. employment* is the ratio of foreign employment compensation to lagged consolidated assets. *Foreign investment* is the ratio of foreign capital expenditures to lagged consolidated assets. *Foreign employment* is the ratio of foreign employment compensation to lagged consolidated assets. *TCJA* is an indicator equalling 1 in 2018 and later. *ForeignCash* is the ratio of a firm's foreign cash in 2017 to the firm's consolidated assets in 2017. *Ownership* is the fraction in 2017 of the firm's stock that is owned by the 5 largest shareholders. Controls are *Age*, $TCJA \times TaxGap$, *Market-to-book*, and *Profitability*, which are defined in Appendix A. Fixed effects are for firm and industry-by-year. Standard errors are clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>U.S. inv.</i> (1)	<i>U.S. emp.</i> (2)	<i>For. inv.</i> (3)	<i>For. emp.</i> (4)
$TCJA \times ForeignCash \times Ownership$	−0.199 (0.163)	0.209 (0.467)	0.135* (0.080)	0.331 (0.383)
$TCJA \times ForeignCash$	0.070 (0.053)	−0.055 (0.151)	−0.048* (0.025)	−0.072 (0.112)
$TCJA \times Ownership$	−0.009 (0.005)	−0.019 (0.016)	−0.012* (0.006)	−0.016 (0.019)
Controls	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes
R ²	0.858	0.912	0.829	0.933
Observations	3146	3146	3146	3146

be negative if the response to the freeing of foreign cash is larger for firms with greater organic foreign profits (i.e., lower voluntary profit shifting). Table 8 reports the results for four outcome variables: U.S. investment, U.S. employment, U.S. R&D, and foreign R&D. In addition to the three U.S. real investment variables, we include foreign R&D because profit shifting is related to the location of intangible assets. For all four outcome variables, the estimates for β_1 are not significantly different from zero. This suggests that the investment response is not significantly related to the degree of profit shifting.

Overall, we do not find evidence that the increased internal capital provided by the TCJA led to an increase in real economic activity. In addition, there is no differential effect based on financial constraints,

Table 8

Profit shifting. This table presents results on the role of profit shifting in firms' responses to the Tax Cuts and Jobs Act. *U.S. investment* is the ratio of U.S. capital expenditures to lagged consolidated assets. *U.S. employment* is the ratio of U.S. employment compensation to lagged consolidated assets. *U.S. R&D* is the ratio of U.S. research and development expenditures to lagged consolidated assets. *Foreign R&D* is the ratio of aggregate foreign research and development expenditures to lagged consolidated assets. *TCJA* is an indicator equalling 1 in 2018 and later. *ForeignCash* is the ratio of a firm's foreign cash in 2017 to the firm's consolidated assets in 2017. *Intangibility* is the four-digit NAICS industry median of the ratio of intangible capital to assets in 2017, where intangible capital is measured following Peters and Taylor (2017). Controls are *Age*, *TCJA* $\times$ *TaxGap*, *Market-to-book*, and *Profitability*, which are defined in Appendix A. Fixed effects are for firm and year. Standard errors are clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>U.S. inv.</i> (1)	<i>U.S. emp.</i> (2)	<i>U.S. R&D</i> (3)	<i>For. R&D</i> (4)
<i>TCJA</i> $\times$ <i>ForeignCash</i> $\times$ <i>Intangibility</i>	0.023 (0.058)	0.024 (0.197)	0.028 (0.068)	0.002 (0.027)
<i>TCJA</i> $\times$ <i>ForeignCash</i>	−0.005 (0.005)	−0.004 (0.010)	−0.001 (0.003)	−0.000 (0.001)
<i>TCJA</i> $\times$ <i>Intangibility</i>	−0.018 (0.041)	−0.021 (0.134)	−0.032 (0.045)	0.001 (0.018)
<i>Intangibility</i>	0.025 (0.020)	−0.036 (0.045)	−0.018 (0.015)	−0.010 (0.017)
Controls	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes
R ²	0.791	0.870	0.918	0.806
Observations	3146	3146	3146	3146

corporate governance, or profit shifting. Taken together, our results are consistent with our sample of firms being relatively unconstrained financially and well-governed.

3.1.2. Mergers and acquisitions

We next consider another form of real economic activity: acquisitions. For reasons similar to those for other real outcomes, increased access to liquidity may lead to increased acquisition activity. This could include decreased cost of capital, as well as exacerbated agency conflicts such as empire building or free cash flow problems (Jensen, 1986; Harford, 1999). Relatedly, Hanlon et al. (2015) finds that trapped foreign cash led to an increase in foreign acquisitions made with cash, and that these acquisitions had lower announcement returns.

We test whether increased access to cash led to an increase in acquisition activity by estimating our baseline Eq. (1) with the dollar volume of acquisitions, scaled by lagged assets, as the outcome variable. The results are reported in Table 9. Similar to Hanlon et al. (2015), we focus on acquisitions made with cash. Consistent with other forms of real activity, we find no effect of increased access to liquidity on firm M&A activity. This is true for both U.S. acquisitions, which are most likely to be affected by the reform, as well as foreign acquisitions. These results contrast with other work that indicates both acquisitions and announcement returns are increasing in foreign cash (see Atwood et al., 2020 and Amberger and Robinson, 2022, respectively). In contrast to existing work, this study uses foreign cash data directly observed by BEA, potentially contributing to the different findings.

In addition to a reduced cost of capital, agency conflicts—such as empire building motives—may lead managers to engage in increased acquisition activity. We consider the role of agency conflicts by including an interaction with our measure of the strength of corporate governance, the equity ownership of the top-5 institutional investors. The results are reported in Table 10. For both U.S. and foreign acquisition activity, we find no evidence that managers facing less institutional oversight responded to the liquidity shock by increasing acquisition activity.

Table 9

Mergers and acquisitions. This table presents estimates of the association between firms' foreign cash and their mergers and acquisitions activity following the TCJA. *U.S. M&A* is the value of cash acquisitions of U.S. targets scaled by lagged consolidated assets. *Foreign M&A* is the value of cash acquisitions of foreign targets scaled by lagged consolidated assets. *TCJA* is an indicator equalling 1 in 2018 and later. *ForeignCash* is the ratio of a firm's foreign cash in 2017 to the firm's consolidated assets in 2017. $\mathbb{I}_{year \leq 2014}$ is an indicator equalling one in 2014 and before. $\mathbb{I}_{year=2015}$ is an indicator equalling one in 2015. $\mathbb{I}_{year=2016}$ through $\mathbb{I}_{year=2019}$ are defined analogously. Controls are *Age*, *TCJA* $\times$ *TaxGap*, *Market-to-book*, and *Profitability*, which are defined in Appendix A. Fixed effects are for firm and industry-year. Standard errors are clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>U.S. M&A</i> (1)	(2)	<i>Foreign M&A</i> (3)	(4)
<i>TCJA</i> $\times$ <i>ForeignCash</i>	−0.016 (0.026)		0.008 (0.007)	
$\mathbb{I}_{year \leq 2014} \times$ <i>ForeignCash</i>		−0.002 (0.030)		−0.004 (0.006)
$\mathbb{I}_{year=2015} \times$ <i>ForeignCash</i>		−0.040 (0.043)		−0.002 (0.012)
$\mathbb{I}_{year=2016} \times$ <i>ForeignCash</i>		0.061 (0.054)		−0.005 (0.007)
$\mathbb{I}_{year=2018} \times$ <i>ForeignCash</i>		−0.003 (0.033)		0.004 (0.009)
$\mathbb{I}_{year=2019} \times$ <i>ForeignCash</i>		−0.033 (0.036)		0.010 (0.009)
Controls	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes
R ²	0.347	0.349	0.348	0.348
Observations	3146	3146	3146	3146

Table 10

Mergers and acquisitions and agency. This table presents results on the role of agency conflicts in mediating the association between firms' foreign cash and their mergers and acquisitions activity following the TCJA. *U.S. M&A* is the value of cash acquisitions of U.S. targets scaled by lagged consolidated assets. *Foreign M&A* is the value of cash acquisitions of foreign targets scaled by lagged consolidated assets. *TCJA* is an indicator equalling 1 in 2018 and later. *ForeignCash* is the ratio of a firm's foreign cash in 2017 to the firm's consolidated assets in 2017. *Ownership* is the fraction in 2017 of the firm's stock that is owned by the 5 largest shareholders. Controls are *Age*, *TCJA* $\times$ *TaxGap*, *Market-to-book*, and *Profitability*, which are defined in Appendix A. Fixed effects are for firm and industry-year. Standard errors are clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>U.S. M&A</i> (1)	<i>Foreign M&A</i> (2)
<i>TCJA</i> $\times$ <i>ForeignCash</i> $\times$ <i>Ownership</i>	−0.056 (0.253)	0.040 (0.059)
<i>TCJA</i> $\times$ <i>ForeignCash</i>	−0.001 (0.083)	−0.004 (0.018)
<i>TCJA</i> $\times$ <i>Ownership</i>	−0.008 (0.024)	−0.003 (0.006)
Controls	Yes	Yes
Fixed effects	Yes	Yes
R ²	0.348	0.347
Observations	3146	3146

3.2. Cash holdings, shareholder payouts, and leverage

In the previous section, we found that firms did not adjust their real economic activity in response to increased liquidity following the TCJA. In this section, we turn to the financial decisions of the firm. How did firms adjust their financial positions in response to a shock to internal liquidity?

To explore how firms responded to the freeing of their foreign cash, we estimate Eqs. (1) and (2) for several financial outcome variables. The results are reported in Table 11. Column (2) shows the dynamics of the consolidated cash holdings as a function of 2017 foreign cash (*ForeignCash*); both variables are scaled by total assets. As expected,

Table 11

Financial response. This table presents estimates of the association between firms' foreign cash and their financial activity. *Cash* is the ratio of consolidated cash to consolidated assets. *Dividends* is the ratio of dividends to shareholders to lagged consolidated assets. *Repurchases* is the ratio of stock repurchases to lagged consolidated assets. *Payouts* is the ratio of the sum of dividends to shareholders and stock repurchases to lagged consolidated assets. *Leverage* is the sum of long term debt and debt in current liabilities to consolidated assets. *TCJA* is an indicator equalling 1 in 2018 and later. *ForeignCash* is the ratio of a firm's foreign cash in 2017 to the firm's consolidated assets in 2017. $\mathbb{I}_{year \leq 2014}$ is an indicator equalling one in 2014 and before. $\mathbb{I}_{year=2015}$ is an indicator equalling one in 2015. $\mathbb{I}_{year=2016}$ through $\mathbb{I}_{year=2019}$ are defined analogously. Controls are *Age* and *TCJA* $\times$ *TaxGap*, which are defined in Appendix A. Fixed effects are for firm and industry-year. Standard errors are clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>Cash</i>		<i>Dividends</i>		<i>Repurchases</i>		<i>Payouts</i>		<i>Leverage</i>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>TCJA</i> $\times$ <i>ForeignCash</i>	-0.110 (0.092)		-0.002 (0.017)		0.151** (0.070)		0.138* (0.076)		0.073 (0.107)	
$\mathbb{I}_{year \leq 2014} \times \text{ForeignCash}$		-0.377*** (0.115)		0.013 (0.030)		-0.088 (0.068)		-0.054 (0.086)		-0.099 (0.156)
$\mathbb{I}_{year=2015} \times \text{ForeignCash}$		-0.154 (0.111)		-0.002 (0.011)		-0.063 (0.086)		-0.070 (0.097)		-0.045 (0.111)
$\mathbb{I}_{year=2016} \times \text{ForeignCash}$		-0.115 (0.070)		0.009 (0.009)		-0.029 (0.088)		-0.018 (0.101)		0.100 (0.072)
$\mathbb{I}_{year=2018} \times \text{ForeignCash}$		-0.248*** (0.082)		0.003 (0.008)		0.091 (0.092)		0.094 (0.100)		-0.003 (0.091)
$\mathbb{I}_{year=2019} \times \text{ForeignCash}$		-0.478*** (0.125)		0.008 (0.017)		0.113 (0.084)		0.117 (0.093)		0.139 (0.148)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.892	0.894	0.856	0.857	0.700	0.701	0.744	0.745	0.893	0.893
Observations	3146	3146	3146	3146	3146	3146	3146	3146	3146	3146

firms with higher 2017 foreign cash experience a larger run up in consolidated cash prior to 2017, as indicated by the positive trend in the estimated coefficients during the pre-period.²³ Following the reform, consolidated cash holdings begin to decline more for those firms with more foreign cash in 2017. For each dollar of foreign cash that was freed by the TCJA, firms reduced consolidated cash holdings by 48 cents two years after the reform. Thus, firms adjusted their consolidated cash positions downward in proportion to the liquidity shock induced by the TCJA. This suggests that at least some of the foreign cash holdings were held for tax purposes, consistent with [Foley et al. \(2007\)](#), [Graham and Leary \(2018\)](#), and [Faulkender et al. \(2019\)](#). It also provides additional support for the assertion that the recent upward trend in corporate cash holdings is linked to the worldwide tax system.

The dynamic estimates in Column (2) indicate that firms that accumulated foreign cash also reduced consolidated cash following the reform (see also Panel E of [Fig. 2](#)). Column (1) reports the single point estimate comparing consolidated cash holdings in the period 2010–17 versus 2018–19. Consistent with the pattern of the dynamic estimates, the point estimate is insignificant, although negative, indicating that the post-reform decline in consolidated cash holdings represents an unwinding of the pre-reform accumulation of foreign cash.

We next test whether firms used this increased access to cash to increase shareholder payouts. Columns (3)–(8) of [Table 11](#) report point and dynamic estimates for dividends, share repurchases, and their sum (total payouts). Share repurchases significantly increase in response to the access to foreign cash, while dividends are unchanged.²⁴ This is

consistent with prior evidence that repurchases (dividends) are more likely used for transitory (persistent) changes to shareholder payouts ([Jagannathan et al., 2000](#); [Guay and Harford, 2000](#)). However, the magnitude of the response in repurchases is relatively modest, at 15 cents per year, for two years, for each marginal dollar of foreign cash. Although modest in magnitude, payouts can account for a substantial fraction of the reduction in consolidated cash holdings.²⁵

To assess the robustness of these results, we repeat the estimates of the dynamic effects for consolidated cash holdings, repurchases, and total payouts, excluding all control variables. The results are reported in Online Appendix Table OA.2. We also explore including the firms' effective tax rates as an additional control variable, reported in Table OA.3. In both cases, the results are very similar.

Our findings on payouts have a notable connection to the public economics literature on dividend taxes. The TCJA reduced the tax cost of accessing foreign cash, including for firms intending to make distributions to shareholders. In this sense, the policy change had an effect similar to a reduction in the dividend tax rate that was enacted as part of the Jobs and Growth Tax Relief Reconciliation Act of 2003 (JGTRRA).²⁶ [Chetty and Saez \(2005\)](#) and [Yagan \(2015\)](#) find that the JGTRRA dividend tax cut is associated with increased dividend payments. Consistent with this result, we find that the reduced tax cost of repatriating foreign cash was associated with an increase in payouts. Furthermore, [Yagan \(2015\)](#) finds that the JGTRRA dividend tax cut had no significant effect on firms' investment and employee compensation. As discussed in Section 3.1, we also find the reduction in the tax cost to firms of accessing their foreign cash had no impact on firms' investment or employment decisions. These complementary results reinforce the notion that dividend taxes impact payout policy, but have minimal effect on firms' real activity.

²³ It is unsurprising that 2017 foreign cash is correlated with pre-TCJA increases in consolidated cash because foreign cash is a component of consolidated cash, the outcome variable (i.e., consolidated cash is the sum of foreign cash and U.S. cash). In other words, we expect—and find—pre-trends in consolidated cash as a function of the treatment in this regression. Therefore, we interpret these results as a description of how firms adjusted their consolidated cash as a function of the foreign cash that was freed. However, the pre-trend in this case works against finding a drop in consolidated cash following reform in that the trend reverses direction immediately post TCJA.

²⁴ [Beyer et al. \(2021\)](#), [Bennett and Wang \(2021\)](#), and [Olson \(2021\)](#) find a similar pattern using alternative empirical approaches.

²⁵ The firm-level response to cash and payouts we find are consistent with the aggregate effects documented by [Darmouni and Mota \(2024\)](#). While we are unable to observe the composition of financial assets in our sample, [Darmouni and Mota \(2024\)](#) find that, in aggregate, significant amounts of corporate bonds were liquidated following the TCJA, coinciding with increased payouts, while the level of cash-like instruments was mostly unchanged.

²⁶ For a survey of the payout policy literature, including the effects of this 2003 dividend tax cut, see [Farre-Mensa et al. \(2014\)](#).

We also consider whether the foreign cash freed by the TCJA was used to pay down debt and reduce leverage. Prior to the TCJA, firms may have used their trapped foreign cash to support borrowing in the U.S.—financing that could be used for U.S. payouts and investment without restriction—as a way to circumvent the repatriation tax and indirectly pay out foreign profits to U.S. shareholders (De Simone and Lester, 2018). A post-reform decline in debt could reflect an unwinding of those pre-TCJA tax-arbitrage positions. However, Columns (9) and (10) of Table 11 show no significant change in book leverage following the TCJA. This result suggests that firms did not use their freed foreign cash to adjust their debt positions.²⁷

We next consider three factors that may affect a firm's financial response to the freeing of its foreign cash. First, we consider whether foreign cash was being held for operational reasons rather than purely for tax reasons. We test this possibility by measuring the TCJA-induced liquidity increase as the portion of foreign cash that is in excess of that required to support foreign operations. Excess foreign cash is determined in four steps. First, the ratio of cash to net property, plant, and equipment is calculated for domestic only Compustat firms (i.e., non-multinationals) in 2017. Second, the median of this ratio within 2-digit NAICS industry bins is matched to each foreign subsidiary and multiplied by the subsidiary's net property, plant, and equipment, to obtain a predicted cash value. Third, this predicted cash value is subtracted from the subsidiary's cash holdings. If the difference is negative it is set to zero. Fourth, these excess cash values are aggregated across a firm's foreign subsidiaries and scaled by consolidated assets. We denote this variable *ExForCash*.

We measure the response in consolidated cash holdings, repurchases, and payouts, by estimating Eq. (2) using the firms' 2017 level of excess foreign cash in place of total foreign cash. Table 12 reports the results. For all three outcome variables, the coefficient estimates using excess foreign cash are very similar to our baseline results. This suggests that the muted response in payouts and consolidated cash holdings was unlikely due to the need to keep the freed cash in the foreign subsidiaries for operational reasons.

Second, managers may receive private benefits from the firm's cash holdings and therefore choose to retain more cash than is optimal from the perspective of shareholders. Indeed, Harford et al. (2017) finds that the value of a firm's foreign cash holdings is discounted by its shareholders, and Kim et al. (2023) finds that shareholders responded positively to a policy that pressured Korean firms to disgorge excess cash. To test whether agency conflicts are associated with higher retention of freed cash, we re-estimate the financial response with the inclusion of an interaction with a proxy for corporate governance. As institutional owners are likely able to exert pressure on management to pay out excess cash, we again use the top-5 institutional ownership of firms as a proxy for strength of governance. Table 13 reports the results of the estimation. In all cases, there is no differential effect attributable to institutional presence, providing no evidence that retention of cash is due to agency conflicts.

Third, we assess whether the low payout of cash is due to a slow adjustment by firms that takes more than two years to complete. We consider this by expanding our post-period estimation window to four years, through 2021. This is possible because the financial responses are based on consolidated data from Compustat, which are available after the BEA sample ends. A caveat is that the expanded sample includes the Covid pandemic that began in 2020, an event that led to a “dash for cash” (Acharya and Steffen, 2020). Therefore, the pandemic presents a significant confounding factor that may slow adjustment and interpreting the results using the extended sample as indicative of long-run outcomes must be done with caution. That said, we do not have a

Table 12

Excess foreign cash. This table presents the financial results with an alternative treatment variable. *Cash* is the ratio of consolidated cash to consolidated assets. *Repurchases* is the ratio of stock repurchases to lagged consolidated assets. *Payouts* is the ratio of the sum of dividends to shareholders and stock repurchases to lagged consolidated assets. $\mathbb{I}_{year \leq 2014}$ is an indicator equalling one in 2014 and before. $\mathbb{I}_{year=2015}$ is an indicator equalling one in 2015. $\mathbb{I}_{year=2016}$ through $\mathbb{I}_{year=2019}$ are defined analogously. *ExForCash* is the ratio of a firm's excess foreign cash (beyond that needed for operations) in 2017 to the firm's consolidated assets in 2017. Appendix A contains a detailed definition. Controls are *Age* and *TCJA × TaxGap*, which are also defined in Appendix A. Fixed effects are for firm and industry-year. Standard errors are clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>Cash</i> (1)	<i>Repurchases</i> (2)	<i>Payouts</i> (3)
$\mathbb{I}_{year \leq 2014} \times ExForCash$	−0.297** (0.040)	−0.098 (0.226)	−0.049 (0.652)
$\mathbb{I}_{year=2015} \times ExForCash$	−0.165 (0.261)	−0.056 (0.623)	−0.071 (0.586)
$\mathbb{I}_{year=2016} \times ExForCash$	−0.121 (0.167)	−0.021 (0.852)	−0.028 (0.831)
$\mathbb{I}_{year=2018} \times ExForCash$	−0.176* (0.082)	0.133 (0.288)	0.127 (0.358)
$\mathbb{I}_{year=2019} \times ExForCash$	−0.434*** (0.004)	0.162 (0.114)	0.161 (0.162)
Controls	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes
R ²	0.893	0.704	0.747
Observations	3146	3146	3146

Table 13

Agency conflicts and financial response. The table presents results on the role of agency conflicts in mediating the association between firms' foreign cash and their financial activity following the TCJA. *Cash* is the ratio of consolidated cash to consolidated assets. *Repurchases* is the ratio of stock repurchases to lagged consolidated assets. *Payouts* is the ratio of the sum of dividends to shareholders and stock repurchases to lagged consolidated assets. *TCJA* is an indicator equalling 1 in 2018 and later. *ForeignCash* is the ratio of a firm's foreign cash in 2017 to the firm's consolidated assets in 2017. *Ownership* is the fraction in 2017 of the firm's stock that is owned by the 5 largest shareholders. Controls are *Age* and *TCJA × TaxGap*, which are defined in Appendix A. Fixed effects are for firm and industry-year. Standard errors are clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>Cash</i> (1)	<i>Repurchases</i> (2)	<i>Payouts</i> (3)
<i>TCJA × ForeignCash × Ownership</i>	0.119 (1.157)	0.348 (0.549)	0.745 (0.614)
<i>TCJA × ForeignCash</i>	−0.147 (0.388)	0.038 (0.183)	−0.102 (0.195)
<i>TCJA × Ownership</i>	0.006 (0.063)	−0.087** (0.042)	−0.112** (0.045)
Controls	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes
R ²	0.892	0.701	0.745
Observations	3146	3146	3146

specific reason to suspect that the differential impact of the pandemic was correlated with firms' 2017 foreign cash holdings.

Table 14 reports the estimation results using the sample extended through 2021. We see that consolidated cash holdings, shown in Column (1), appears mostly unchanged in 2020 and 2021. This is similarly true for repurchases and payouts, shown in Columns (2) and (3). If anything, there is a slight reversal in 2020 and 2021, suggesting that a narrow event window does not explain the muted financial response. The results also support our choice to use a two-year event window.

In summary, we do not find evidence that the payout or cash retention rates are associated with operational needs for cash holdings, agency conflicts, or a delay in response.

²⁷ Carrizosa et al. (2020), Beyer et al. (2021), and Sanati (2022) explore the response in debt to the TCJA, but these studies do not explore the effect of increased liquidity.

Table 14

Financial results with extended sample. This table presents financial results with the sample extended through 2021. *Cash* is the ratio of consolidated cash to consolidated assets. *Repurchases* is the ratio of stock repurchases to lagged consolidated assets. *Payouts* is the ratio of the sum of dividends to shareholders and stock repurchases to lagged consolidated assets. $\mathbb{I}_{year \leq 2014}$ is an indicator equalling one in 2014 and before. $\mathbb{I}_{year=2015}$ is an indicator equalling one in 2015. $\mathbb{I}_{year=2016}$ through $\mathbb{I}_{year=2021}$ are defined analogously. *ForeignCash* is the ratio of a firm's foreign cash in 2017 to the firm's consolidated assets in 2017. Controls are *Age* and *TCJA×TaxGap*, which are defined in Appendix A. Fixed effects are for firm and industry-year. Standard errors are clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>Cash</i> (1)	<i>Repurchases</i> (2)	<i>Payouts</i> (3)
$\mathbb{I}_{year \leq 2014} \times \text{ForeignCash}$	−0.380*** (0.115)	−0.090 (0.069)	−0.056 (0.085)
$\mathbb{I}_{year=2015} \times \text{ForeignCash}$	−0.152 (0.111)	−0.058 (0.087)	−0.066 (0.098)
$\mathbb{I}_{year=2016} \times \text{ForeignCash}$	−0.119* (0.071)	−0.025 (0.088)	−0.013 (0.102)
$\mathbb{I}_{year=2018} \times \text{ForeignCash}$	−0.237*** (0.083)	0.089 (0.093)	0.093 (0.101)
$\mathbb{I}_{year=2019} \times \text{ForeignCash}$	−0.464*** (0.124)	0.111 (0.085)	0.117 (0.093)
$\mathbb{I}_{year=2020} \times \text{ForeignCash}$	−0.371** (0.154)	−0.011 (0.087)	−0.014 (0.096)
$\mathbb{I}_{year=2021} \times \text{ForeignCash}$	−0.433*** (0.153)	−0.063 (0.127)	−0.117 (0.124)
Controls	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes
R ²	0.886	0.698	0.745
Observations	3479	3479	3479

Given these results, a question remains: why was less than one-third of each dollar of new liquidity used to make repurchases, with half of it retained as cash, once it was unlocked by the TCJA? Indeed, conventional wisdom and existing empirical evidence suggest that unconstrained firms would likely respond to the liquidity shock by paying out the excess liquidity to shareholders rather than retaining it as cash (Almeida et al., 2004; Dharmapala et al., 2011; Dharmapala, 2024).

In contrast, our results suggest that firms do not quickly adjust cash ratios in response to positive liquidity shocks, similar to what Graham and Leary (2018) describes as “loose” adherence to cash targets. In other words, nominal cash holdings appear to be somewhat sticky, analogous to the nominal stickiness of debt documented by Welch (2004). One possibility is that this loose cash targeting is the result of fixed costs of adjustment. However, given that paying out excess cash is unlikely to be particularly costly, an alternative explanation is that firms may be roughly indifferent between paying out and retaining the increased liquidity. Specifically, prior to reform, the marginal benefits of retaining cash were high due to repatriation taxes, leading to foreign cash accumulation. After reform, only the “traditional” cash holdings trade-off remains, that is, costs of carry and agency versus precautionary savings motives (e.g., Nikolov and Whited, 2014). For our sample of firms that appear to be well-governed and financially unconstrained, these traditional marginal costs and benefits are each likely quite small, leading to optimal cash holdings that may be roughly indeterminate. Simply put, firms had a strong tax incentive to accumulate cash prior to reform, but little compelling reason to adjust cash downward once the repatriation tax was removed by the TCJA.

It is worth noting that firms did not reduce leverage in conjunction with their high retention of cash after the reform. Gao et al. (2021) show that it may be optimal for firms to hold cash alongside their debt positions because it reduces the likelihood of default and acts as a type of collateral, thus lowering the cost of borrowing. Their model extends

the traditional precautionary savings model, and may help to explain why firms did not pay out more of their freed foreign cash.

It is also possible that the optimal level of internal liquidity—represented by U.S. cash before the reform and consolidated cash after—increased as a result of the TCJA. Specifically, the reduction in the U.S. corporate tax rate, which affected all firms in our sample, likely reduced the tax cost of holding cash inside the firm.²⁸ Reducing the marginal cost of holding cash would, all else equal, increase the optimal level of cash holdings, which is consistent with our results. This implies that the marginal benefits of cash need not be large for firms to choose to retain freed cash.

4. Conclusion

The Tax Cuts and Jobs Act unlocked up to \$1.7 trillion in cash trapped in U.S. multinational firms' foreign operations. We construct a direct measure of firms' exposure to this policy change using detailed data from the Bureau of Economic Analysis and implement a difference-in-differences research design to explore its impact on their real and financial policies in the U.S. and abroad. We find no significant association between the policy change and firms' capital expenditures, employment, research and development, or mergers and acquisitions at either their domestic or foreign operations. These findings do not vary with the degree of firms' financial constraints. In terms of financial outcomes, the policy shock is not associated with firms' dividend or leverage decisions, but has a moderate positive association with stock repurchases. Firms with more trapped liquidity pre-TCJA also reduce their consolidated cash holdings post-TCJA by 48 cents for each dollar of foreign cash unlocked. Firms thus retain about half of the freed cash. Strength of corporate governance does not appear to mediate firms' financial responses to the policy change. Our findings suggest that increased access to capital likely did not have a significant effect on firms' real decisions, and demonstrates only a loose targeting of cash ratios. This is consistent with small marginal costs and benefits of holding cash for these firms, as well as a limited incentive for firms to pay out excess cash accumulated overseas.

CRedit authorship contribution statement

James F. Albertus: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Brent Glover:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Oliver Levine:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

²⁸ Cash holdings face a tax penalty when held in an interest-bearing account inside the firm if the corporate tax rate exceeds the personal tax rate. For example, this is the marginal cost used in the precautionary savings model of Riddick and Whited (2009) in order to bound cash holdings. Lowering the corporate tax rate lowers this penalty.

Appendix A. Variable definitions

We winsorize each variable at the 1% and 99% thresholds of its empirical distribution to mitigate the influence of outliers. All financial variables are denominated in millions of 2018 U.S. dollars.

- *Age* is the logarithm of 1 plus the number of years since the firm first appears in the Compustat data.
- *Cash* is the ratio of consolidated cash to consolidated assets.
- *Dividends* is the ratio of dividends to shareholders to lagged consolidated assets.
- *EffectiveTaxRate* is the ratio of consolidated taxes paid to consolidated pretax income.
- *ExForCash* is the ratio of consolidated excess foreign cash in 2017 to consolidated assets in 2017. Excess foreign cash is determined in four steps. First, the ratio of cash to net property, plant, and equipment is calculated for domestic only Compustat firms (i.e., non-multinationals) in 2017. Second, the median of this ratio within 2-digit NAICS industry bins is matched to each foreign subsidiary and multiplied by the subsidiary's net property, plant, and equipment, to obtain a predicted cash value. Third, this predicted cash value is subtracted from the subsidiary's cash holdings. If the difference is negative it is set to zero. Fourth, these excess cash values are aggregated across a firm's foreign subsidiaries and scaled by consolidated assets.
- *ForeignCash* is the ratio of aggregate foreign cash in 2017 to consolidated assets in 2017.
- *Foreign employment* is the ratio of aggregate foreign employment compensation to lagged consolidated assets.
- *Foreign investment* is the ratio of aggregate foreign capital expenditures to lagged consolidated assets.
- *Foreign M&A* is the value of cash acquisitions of foreign targets scaled by lagged consolidated assets.
- *Foreign R&D* is the ratio of aggregate foreign research and development expenditures to lagged consolidated assets.
- *HP* is the 2017 value of the Hadlock Pierce financial constraint measure as calculated in the last paragraph of Section 3.2 of Hadlock and Pierce (2010).
- $\mathbb{I}_{Year \leq 2014}$ is an indicator equalling one in 2014 and before.
- $\mathbb{I}_{Year=2015}$ is an indicator equalling one in 2015. $\mathbb{I}_{Year=2016}$ through $\mathbb{I}_{Year=2021}$ are defined analogously.
- *Intangibility* is calculated in three steps. First, we calculate the ratio of intangible capital to consolidated assets in 2017 for each firm. We use the data on intangible capital calculated by Peters and Taylor (2017). Second, we calculate the four-digit NAICS industry medians of these ratios. Third, we match the industry median ratio back to the firm-year panel.
- *Leverage* is the sum of long term debt and debt in current liabilities to consolidated assets.
- *Market-to-book* is the ratio of market value of assets to book value of assets. Market value of assets is consolidated assets plus market value of equity less book value of equity. Market value of equity is the product of shares outstanding and share price. Book value of assets is consolidated assets.
- *Ownership* is the fraction in 2017 of a firm's equity ownership attributable to its top 5 institutional investors, as defined by Ferreira and Matos (2008).
- *Payouts* is the ratio of the sum of dividends to shareholders and stock repurchases to lagged consolidated assets.
- *Profitability* is the ratio of consolidated net income to consolidated assets.
- *Repatriations* is the ratio of repatriations to lagged consolidated assets. Specifically, repatriations are dividends from affiliates reported on the Activities of U.S. Multinational Enterprises surveys that are 100% directly-owned by their U.S. parent.

- *Repurchases* is the ratio of stock repurchases to lagged consolidated assets.
- *TaxGap* is the U.S. statutory tax rate in 2017 less the sales weighted average foreign statutory tax rate in 2017.
- *TCJA* is an indicator equalling one in 2018 and later.
- *U.S. employment* is the ratio of U.S. employment compensation to lagged consolidated assets.
- *U.S. investment* is the ratio of U.S. capital expenditures to lagged consolidated assets.
- *U.S. M&A* is the value of cash acquisitions of U.S. targets scaled by lagged consolidated assets.
- *U.S. R&D* is the ratio of U.S. research and development expenditures to lagged consolidated assets.

Appendix B. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jfineco.2025.104006>.

Data availability

MendeleyData Packet for "The real and financial effects of internal liquidity: Evidence from the Tax Cuts and Jobs Act" (Original data)

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